

1. Answer **only one** of the following as **assigned** in the given Google Sheet:
Prove that the following language is not regular using Pumping Lemma:

a. $\{a^{n^4} \mid n > 0\}$

b. $\{a^{n+n^2} \mid n > 0\}$

c. $\{a^{2n}b^{n^2} \mid n > 0\}$